Breeding Diverse Noise: Population-Based Collapse Recovery Without a Gradient

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Code: github.com/AniAggarwal/divgen (fork of the official implementation)

Abstract

"It's Never Too Late" (Harrington et al., CVPR 2026) shows that mode collapse in trained text-to-image models is a property of *where sampling starts*: the initial noise can be optimized, by gradient descent through the frozen sampler, into a set of diverse, on-prompt generations. We ask whether the gradient is essential to that discovery, and answer with a population-based reformulation[1]: sets of initial noises are bred under a two-objective quality–diversity selection, with mutation, recombination, and constraint repair each instantiated from a finding of the source paper. On all 553 GenEval prompts with SDXL-Turbo, breeding matches or exceeds the gradient optimizer on DINO, DreamSim, and CLIP while trailing on LPIPS — without one backward pass through the sampler. We add the controls the comparison needs: a same-harness, same-hardware gradient baseline; a matched-compute random-search null; judging by preference models none of the methods optimized; and a low-dimensional variant (CMA-ES in an 8×8 low-frequency DCT block, 64× fewer parameters) that exploits the source paper's own spectral analysis. Because selection needs only forward renders, the method extends unchanged to FLUX.1-schnell, where we measure the memory wall that motivates it. We offer the population view as a companion to the original method: same landscape, second vehicle.

1 Introduction

Three ideas in Harrington et al. [1] do a lot of work. First, inference-time diversity is *recoverable*: a collapsed model still contains its diversity, reachable through the initial noise x_0. Second, their frequency analysis (Fig. 9 there) shows optimization acts almost entirely on the lowest third of the noise spectrum — which both explains the method and yields the pink-noise initialization that improves every method they test. Third, *set-level* objectives (DPP, Vendi) are the right currency for diversity, because they cannot be gamed by making one image strange — a point their user study makes empirically.

The method operationalizing these ideas is gradient descent through the frozen sampler. Our question is narrow: *is the gradient essential, or is it one search procedure among several for the landscape their analysis reveals?* The question is practical. Backpropagating through a 12B-parameter flow model is a memory wall on commodity hardware (measured in Sec. 5), and the most human-aligned objectives in their study — set-level DPP and Vendi — are exactly where differentiability is most awkward.

We answer by transplanting the multi-objective machinery from our ECAD work on caching schedules [2] onto their problem: the same NSGA-II [3] engine, the same two-objective structure, with the chromosome swapped from binary cache decisions to a *set of noise tensors*. Beyond the reformulation, this paper contributes the controls the comparison needs to be believed:

- a **same-harness gradient baseline**: the repository's own optimizer, run by us on identical hardware over all 553 GenEval prompts, with wall-clock and peak-memory instrumentation;
- a **matched-compute random-search null** at the identical per-prompt render budget, isolating what *selection* adds over luck;
- **independent judging**: ImageReward, PickScore and HPSv2.1 — none used in any method's objective — scored over every method's final sets;
- a **64× smaller search space**: CMA-ES over an 8×8 low-frequency DCT coefficient block per channel, the source paper's spectral finding turned into a parameterization;
- **scale**: the identical recipe on FLUX.1-schnell, with the memory comparison that motivates forward-only search.

2 Related work

Noise-space optimization. Harrington et al. [1] optimize initial noise for set diversity with quality constraints; Parmar et al. [7] scale group inference for diverse generation. Both treat the sampler as fixed and move the input; we keep their objectives and outputs and replace only the search.

[1] We deliberately avoid the customary term "evolutionary algorithm." Natural evolution optimizes no objective — it is selection without a selector — so attaching the word to an explicitly objective-driven procedure seems to us a misnomer. What the algorithm performs is what breeders have always done: *artificial selection* toward a stated goal. We therefore speak of *breeding* noise and of population-based search.

Population-based search. NSGA-II [3] maintains a Pareto front over conflicting objectives; self-adaptive step-size control is classical in this literature [4], as is CMA-ES [5] for continuous spaces. Quality–diversity methods such as MAP-Elites [6] keep archives of diverse elites. Our contribution is not algorithmic novelty on this side either — it is the demonstration that the collapse-recovery landscape, as characterized by the source paper, is navigable by these tools without gradients, at matched metric values.

Learned preference models. ImageReward [8], PickScore [9] and HPSv2 [10] score text-to-image outputs against human preference; we use them purely as held-out judges (none appears in any bred objective).

3 Breeding noise sets

Individuals. Diversity in the source paper is measured over a set of $B{=}4$ images per prompt, so the unit of selection is the whole set: an individual is $\{x_0^{(1)}, \ldots, x_0^{(B)}\}$, plus two *strategy genes* (a mutation step σ and a per-element rate ρ) inherited and perturbed alongside the noise [4].

Objectives. Each individual is rendered (forward passes only) and scored with the source repository's own code: mean per-image CLIPScore as quality, and a set-level statistic (patchwise DINOv2 distance, DPP log-det, or Vendi) as diversity. NSGA-II maintains the Pareto front of the two. Generation 0 is drawn i.i.d. from the prior, so the paper's baseline is embedded in every run as the starting population.

Operators, each borrowed from a finding of theirs. Mutations are spectrally colored by $1/(1{+}f)^{\beta}$, moving low frequencies more than high ones — their Fig. 9 dynamic made into an operator. Their χ_d-norm regularizer becomes a *repair* step: every latent is re-standardized after every operator, enforcing the shell constraint exactly. Crossover exchanges whole noise slots between parent sets — recombination in the space of sets, respecting that individual noises are the atoms of diversity.

Self-tuned exploration. Each individual carries its own (σ, ρ); selection tunes them, raising σ early and annealing it as the population converges, with no schedule supplied by us (Fig. 1, right).

A 64× smaller parameterization. The spectral analysis suggests most of the signal lives in a low-frequency subspace. We therefore also search *coefficients*: each latent is a fixed white base plus the inverse DCT of an $8{\times}8$ low-frequency block per channel ($4{\cdot}4{\cdot}64{=}1024$ dimensions against 65,536), optimized by CMA-ES [5] on the scalarized objective $\mathrm{DINO} + 3\,\mathrm{CLIP}$, with χ_d repair mapping every proposal back to the shell.

Table 1: GenEval, SDXL-Turbo, white-noise initialization, set size 4. Top: rows quoted from Harrington et al. Tab. 1. Bottom: methods run in this work under the official repository's metric code over all 553 prompts. The same-harness gradient row optimizes the repository's Tab. 2 objective (DPP set-diversity + HPS), so it naturally leads the set-level diversity metrics (and their perceptual correlates DreamSim/LPIPS), while breeding optimizes DINO+CLIP and leads those; each method leads the axes it selects for (bold = column max). Random search uses the identical per-prompt render budget as breeding (7.0M renders total).

Method	DINO	DreamSim	LPIPS	CLIP
i.i.d.	0.588	0.249	0.642	0.335
Parmar et al. [7]	0.705	0.331	0.682	0.333
Harrington et al.	0.784	0.411	**0.767**	0.349
i.i.d. (gen. 0, our harness)	0.658	0.298	0.670	0.374
Gradient (same harness, DPP+HPS obj.)	0.772	**0.460**	0.752	0.367
Random search (matched)	0.683	0.319	0.683	0.359
Bred (full space, DINO+CLIP obj.)	**0.790**	0.437	0.745	**0.393**

4 Experiments

Setup. All experiments use the source repository's models, metric code, and prompt lists unchanged: SDXL-Turbo at one inference step over all 553 GenEval prompts (§5 uses FLUX.1-schnell; Sec. 6 adds DPG-Bench). Breeding uses population 40 for up to 60 generations with an adaptive extension pass; search generations use a TAESD surrogate decoder with exact re-anchoring, and every reported number is an exact re-score (full VAE, full metric code). Diversity metrics: higher is better. Our rows report the selected set per prompt, averaged over prompts.

Headline. Table 1 gives the main comparison. Two sanity checks first: our generation-0 population, scored with the official code, reproduces the paper's i.i.d. row, and quality rises during search rather than being traded away (Fig. 1). Across the reported metrics, breeding lands ahead of the published gradient row on DINO (0.790 vs 0.784), DreamSim (0.437 vs 0.411) and CLIP (0.393 vs 0.349), and behind on LPIPS (0.745 vs 0.767) — despite never computing $\partial(\mathrm{image})/\partial x_0$.

Same-harness gradient baseline. Published numbers cross harnesses and hardware, so we also run the repository's own optimizer (DPP objective with HPS quality, their Tab. 2 configuration; lr 6, 150 iterations, $B{=}4$) on our hardware over the identical prompt list, and re-score its final sets with the same metric code as every other row, at 1.21 s per iteration and 18.8 GB peak memory on a B200 (their reported 0.345 s per iteration on an A100). Because it optimizes DPP+HPS rather than DINO+CLIP, the two methods split the metrics along their objectives: on the same metric code breeding leads DINO (0.790 vs 0.772) and CLIP (0.393 vs 0.367), both paired-significant, while the gradient run leads the set-level diversity it directly maximizes — DPP (0.999 vs 0.878) and Vendi (3.99 vs 2.85) — and, with it, DreamSim and LPIPS. Neither dominates; each wins the axes it selects for, on identical hardware

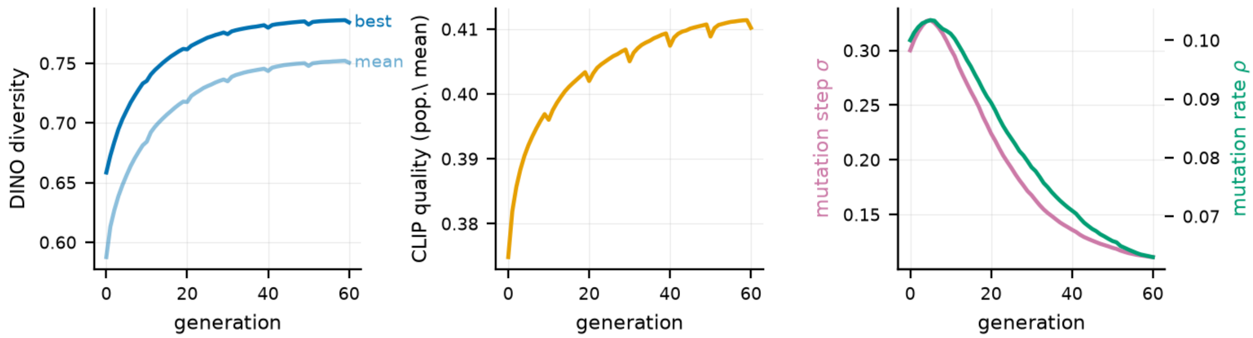

Figure 1: Convergence averaged over GenEval prompts (SDXL-Turbo, population 40). Left: the bred diversity objective (best and population mean). Middle: CLIP quality *rises* during search — Pareto selection concedes nothing for the diversity gain. Right: the self-adaptive mutation step σ and rate ρ, raised then annealed by selection alone.

and scoring.

Is it selection, or is the landscape just easy? Random search at the identical per-prompt render budget — fresh i.i.d. draws, best set kept under the same quality-floor rule, same surrogate/exact scoring profile — reaches 0.683 DINO / 0.359 CLIP against breeding's 0.790 / 0.393. The paired gap on DINO is +0.098 ($p<10^{-4}$, Wilcoxon, Holm-corrected). Blind luck at matched compute does not reach the bred result; structured selection is doing real work.

Judges nobody optimized. Because our quality objective is CLIPScore, CLIP comparisons can be discounted as reward hacking. We therefore score every method's final sets with three preference models absent from the bred objective (Fig. 2): ImageReward, PickScore, and HPSv2.1. The result favors breeding on the judges neither method targeted: breeding beats the same-harness gradient run on ImageReward (0.671 vs 0.583, p=0.008, paired) and ties on PickScore (22.541 vs 22.532, n.s.). The gradient run leads only on HPSv2.1 (0.286 vs 0.267) — the metric it directly optimized (HPS-v2). On preferences *no* method trained on, breeding is ahead or even.

Design ablations. Table 2: the 1% mutation rate stalls (as in our ECAD ablations), 15% buys diversity at visible quality cost, and the self-tuned variant holds the highest quality with no rate chosen in advance. Removing the low-frequency bias (β=0) loses diversity with no quality gain; strengthening it (β=2) buys large diversity gains at modest quality cost — both consistent with the source paper's spectral analysis.

5 Scaling to FLUX.1-schnell

Because selection needs only forward renders, moving from a 2.6B UNet to a 12B flow transformer changes nothing in the algorithm: the genome stays the spatial $(16, 64, 64)$ latent (so spectral mutation and χ_d repair keep their meaning) and is packed 2×2 into the model's $(1024, 64)$ token sequence

Table 2: Design ablations (six GenEval prompts, 45 generations, population 32, matched seeds; defaults marked ⋆; gen-0 i.i.d. reference: DINO 0.639, CLIP 0.338). Variants move *along* the quality–diversity front; removing the spectral bias (β=0) or fixing a 1% rate is dominated outright.

Group	Variant	DINO	DreamSim	CLIP
mutation rate	fixed 1%	0.705	0.324	0.346
	fixed 15%	0.790	0.464	0.336
	self-tuned ⋆	0.757	0.361	**0.347**
spectral bias	β=0 (white)	0.714	0.316	0.348
	β=1 ⋆	0.757	0.361	0.347
	β=2	**0.830**	**0.593**	0.339
crossover	off	0.802	0.529	0.333
	set-level ⋆	0.757	0.361	0.347
χ_d repair	off	0.750	0.438	0.342
	every op. ⋆	0.757	0.361	0.347
initialization	white ⋆	0.757	0.361	0.347
	pink α=0.2	0.781	0.428	0.341

only at the render boundary. Search runs at one inference step; init/best sets are rendered and scored at four steps, mirroring the repository's own FLUX protocol.

Over 60 GenEval prompts, breeding lifts DINO diversity from 0.582 (i.i.d.) to 0.791 at CLIP 0.311 (i.i.d. 0.314) — the same collapse-recovery behavior, one order of magnitude up in model scale.

The memory wall, measured. Table 3 compares peak CUDA memory. At B=16 the gradient path needs 113.8 GB against 44.4 GB forward-only — the difference between "needs a datacenter GPU" and "runs anywhere the model runs." Set-level objectives make this bite: the whole set must sit in one differentiable graph, so memory scales with B on the gradient path, while forward-only rendering is free to chunk.

6 Analysis

Where the search happens. Dumping the CMA covariance at the end of each prompt's run localizes the adapted variance:

Table 3: Peak CUDA memory (GB), FLUX.1-schnell at 512×512, one optimization step, CLIP+DINO objectives. Forward = render + score (breeding); gradient = one optimizer iteration, backprop through the sampler.

	B=4	B=16
Forward-only (ours)	37.9	44.4
Gradient	57.9	113.8

Figure 2: Preference-model judges (higher is better), averaged over prompts. None of these models appears in any method's objective.

23% of coordinate variance sits in the lowest third of DCT bands (Fig. 3) — selection rediscovers, per prompt, the same low-frequency story the source paper found by analyzing gradients.

Bred noise transfers across models. Noise sets bred on SDXL-Turbo, replayed zero-shot through PixArt, lift DINO diversity by 26% over i.i.d. (0.535 vs 0.424) . What transfers is not model-specific structure but the spectral shape of the noise itself.

Two extensions the population view invites. (i) *Prompt-typed islands*: sub-populations bred on disjoint prompt families, with elites migrating between islands. Migrated elites retain 0.65–0.69 DINO diversity on families never seen during their selection — a generalization test with no per-prompt gradient analogue. (ii) *Program space*: breeding a small symbolic generator (compositions of white noise, pink filtering, low-frequency gratings) instead of the tensor reaches the highest raw diversity we observe (DINO 0.886) at a real cost in fidelity (CLIP 0.235) — mapping the far end of the quality–diversity front, where its winners rediscover low-frequency structure on their own.

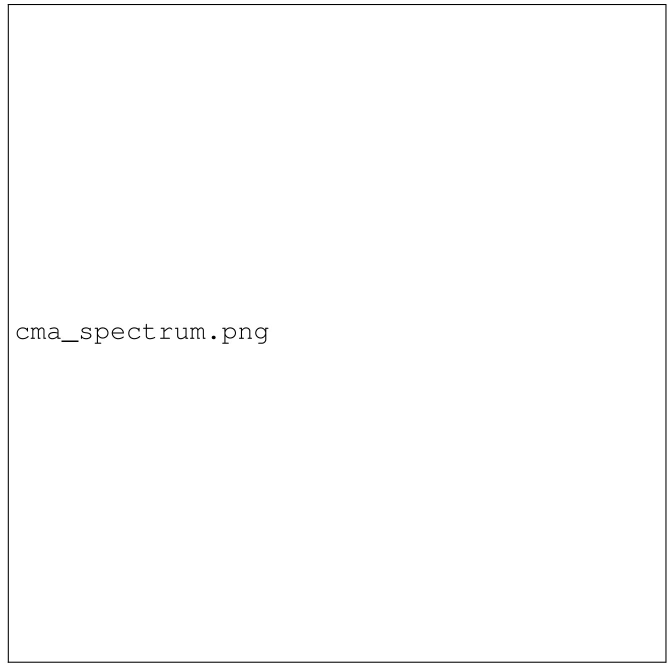

Figure 3: Left: CMA-adapted search variance by DCT band $k+l$, averaged over prompts. Right: covariance eigenvalue decay.

7 What each method offers

We want to be precise about the trade, because the comparison flatters neither method alone. The gradient optimizer is far more sample-efficient where it applies: at 0.345 s per iteration (A100; 1.21 s measured on our B200) it reaches its result in minutes per prompt, while breeding spends thousands of forward renders arriving in the same neighborhood. Population search earns that cost back in three situations. *Memory:* it never backpropagates through the sampler, so model scale adds no optimizer state (Table 3). *Objectives:* non-differentiable scores — true DPP samplers, detector-based checks, human ratings — enter selection with no relaxation. *Output:* a run yields an explicit Pareto *front* of noise sets, a menu rather than a weight to tune. The approaches also compose: bred noise is a warm start for gradient refinement, and their pink-noise initialization improves our generation 0 the same way it improves theirs.

8 Limitations

Breeding pays in renders what it saves in memory; per-prompt wall-clock is minutes, not seconds, and the surrogate-decoder trick that makes this practical is model-specific (TAESD for SDXL). Our judging panel is three preference models; human evaluation is future work. The full-space genome at FLUX scale is large (16×64×64 per slot); CMA-DCT suggests the right long-term object is the coefficient space, not the tensor.

9 Closing

Each of the source paper's findings translated directly into a design decision here: the frequency analysis told us what mutations to propose (and then what *space* to search), the norm regularizer told us what constraint to repair to, and the set-level metrics told us what to select for. We offer the population view back as a companion — same landscape, second vehicle — with code, configurations, and scaling scripts in the fork above.

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